Multitarget Detection Algorithms for Multitemporal Remote Sensing Data

Yanxin Xi, Luyan Ji, Weitun Yang, Xiurui Geng, and Yongchao Zhao

***Abstract*— Target detection is always an important topic in the field of hyper/multispectral remote sensing image processing. At present, target detection algorithms in this field are generally limited to processing single-temporal remote sensing data, and they cannot obtain satisfactory results when the spectra of target and background are similar to each other. Recently, a target detection algorithm called filter tensor analysis (FTA), which is specially designed for multitemporal remote sensing data, has been reported and has achieved better detection results in many cases than the traditional single-temporal methods. However, FTA can only extract one target of interest at a time, and it cannot work when there are multiple targets of interest in the image. Therefore, considering that the matrix form of the FTA method is similar to that of the constrained energy minimization (CEM) model, it naturally comes to us that we can combine the tensor filter in FTA and the multiple target constraints to detect multiple targets by fully exploiting the time-series information in multitemporal data. To be specific, through: 1) adding the "output to one" constraints to the multiple targets in FTA; 2) applying linear/nonlinear function to the outputs of FTA for the multiple targets; and 3) modifying the autocorrelation matrix in FTA, four multitarget detection algorithms for multitemporal remote sensing data are proposed in this article. Experiments with simulation data and real data both show the effectiveness and superiority of the proposed methods.**

***Index Terms*— Filter tensor analysis (FTA), multitarget detection, multitemporal, remote sensing data.**

I. Introduction

REMOTE sensing technology makes more and more remote sensing data easily available, which have high spatial resolution and temporal resolution. Among a wide range of utilities of the remote sensing images [1]–[3], target detection has always been very important [4]–[6]. Therefore, target detection has received much attention and has been the research hotspot in remote sensing in the past decades [7]–[9]. It takes a key role in many applications, such as camouflage object detection, disaster control, land planning, urban monitoring, and target tracking [10]–[14]. Theoretically, target detection in remote sensing images is actually a dichotomy problem: marking pixels as the target or background.

Mainly, the target detection in remote sensing images can be conducted spatially or spectrally [15]. In the spatial domain, the detection is usually applied to optical remote sensing images, especially the images with high spatial resolution. The traditional target detection methods in optical remote sensing images can be divided into four separate parts [16]: template matching-based method, knowledge-based method, object-based image analysis-based method, and machine learning-based method. The template matching-based methods include the rigid template matching [17], [18] and the deformable template matching [19], [20]. The knowledge-based methods can be divided into geometric knowledge [21] and context knowledge-based methods [22]. The object-based image analysis-based methods include image segmentation and object classification [23], [24]. At last, machine learning-based methods are most widely used in recent years, which contains feature extraction [25], [26] and classifier training methods [27]–[33].

In this article, we mainly focus on the target detection algorithms using spectral information. In hyper/multispectral remote sensing images, the spectral difference between the target and background is the physical basis for effective target detection. In target detection in the spectral domain, the objects in the remote sensing images whose ground sampling distance is generally larger than the size of targets can be detected effectively. Many target detection algorithms in the spectral domain have been proposed recently [34]. Most of those algorithms come from the field of hyper/multispectral images and are based on the statistics of the input images. The common algorithms include the matched filter (MF) [34], the adaptive cosine estimator (ACE) [35], [36], the constrained energy minimization (CEM) [7], [15], [36], [37], the clever eye (CE) [38], and some nonlinear detectors dealing with complex scenarios, such as the kernel CEM [39], the kernel MF [40], and the piecewise linear strategy (PLS) [41]. MF and ACE consider the target detection as a hypothesis test problem. CEM, which only needs the knowledge of the desired target, designs a finite impulse response (FIR) filter in a way that the output energy of the filter is minimized, while the filter's response to the target signature is unity [37]. CE algorithm automatically finds the best data origin and moves the data cloud to find the optimal direction to project the data. As a result, CE has lower output energy than CEM and MF [38]. There are other

Manuscript received November 11, 2019; revised April 23, 2020 and October 11, 2020; accepted November 5, 2020. Date of publication November 25, 2020; date of current version December 2, 2021. This work was supported in part by the Second Tibetan Plateau Scientific Expedition and Research Program (STEP) under Grant 2019QZKK0206, and in part by the Chinese National Natural Science Foundation under Grant 61701477. *(Corresponding author: Luyan Ji.)*

The authors are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China, and also with the Key Laboratory of Technology in Geo-Spatial Information Processing and Application System, Chinese Academy of Sciences, Beijing 100190, China (email: jily@mail.ustc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2020.3037087

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

algorithms recently derived from CEM and ACE to further improve the performance of the detection algorithms in a complicated background. Yang and Shi [42] proposed a sparse CEM and sparse ACE algorithms, which uses the l_1-norm regularization term to restrict the output to be sparse, because the target usually occupies a few pixels in real remote sensing images. Geng *et al.* [43] also designed the augmented-CEM (ACEM), and they proved that ACEM is mathematically equivalent to MF. Zou and Shi [44] proposed the hierarchical CEM (hCEM) to suppress the background while preserving the target spectra. Considering the situation that most existing methods put a strong constraint on the target spectrum and that brings more difficulties in detecting the target accurately, Yang *et al.* [45] also proposed an algorithm by employing an inequality constraint. Wang *et al.* [46] introduced a two-time detection scheme of using the principle component CEM and matrix taper CEM simultaneously. They [47] also put forward the sparse weighted CEM (SWCEM) by combining CEM with the sparse representation. Zhao *et al.* [48] designed the Ensemble-based cascaded CEM (E-CEM), including the strategies of "cascaded detection," "random averaging," and "multiscale scanning" to improve the detection accuracy. Apart from the abovementioned methods derived from CEM, there are also some hyperspectral target detection methods based on machine learning techniques. Du and Li [49] applied the subtraction pixel pair features to the hyperspectral target detection framework. Xie *et al.* [50] imposed the spectral regularization into the unsupervised networks to extract intrinsic properties underlying hyperspectral images in order to achieve effective target detection. Shi *et al.* [51] accomplished the hyperspectral target detection with region-of-interest feature transformation and multiscale spectral attention. They [52], [53] also utilized the macro-micro features extracted by a 3-D residual autoencoder and discriminative feature learning with distance constrained stacked sparse autoencoder to effectively detect targets in hyperspectral imagery, respectively. Even though those methods mentioned earlier have achieved great success, they are concerned only with one kind of spectrum in general, which we call the single-target detection problem.

In most hyper/multispectral remote sensing data that include multiple targets or single target with multiple spectral characteristics, detectors for single-target spectrum have very limited performance. For resolving the problem, Ren *et al.* [54] designed several multiple-target detection methods on the basis of CEM, i.e., multiple-target CEM (MTCEM), Sum CEM (SCEM), and Winner-Take-All CEM (WTACEM). However, those methods are all designed for single-temporal data. Under the circumstances that the spectra of targets and background are very similar in a given time phase, identifying the targets would be difficult for the algorithms mentioned earlier using single-temporal data. Those algorithms cannot be applied to the multitemporal remote sensing data directly. Though they can be applied to each single time phase of the multitemporal data, how to combine all the single-temporal outputs would be a problem.

With various types of multitemporal remote sensing data being available, e.g., Landsat and Moderate Resolution Imaging Spectroradiometer, effective applications of multitemporal remote sensing data have been studied [55]–[58]. Aimed for the target detection for multitemporal hyper/multispectral remote sensing data, the filter tensor analysis (FTA) [59] is proposed based on the concept of multilinear function. It designs a tensor filter to minimize the total output energy while keeping the target output equal to one. When the data of less-correlated time phases are selected, FTA detects the target very well. Yet, FTA is only applicable to single-target detection in multitemporal data. When we have to detect several target spectra simultaneously in a hyper/multispectral remote sensing data, multitarget detection algorithms for multitemporal remote sensing data are needed intensely.

Hence, in this article, we put forward several multitarget detection algorithms for multitemporal multispectral data according to the following ideas. The performance of FTA is sensitive to the selection of the target signature. To alleviate this influence: 1) we consider multiple target signatures in the filter design and 2) we also use a linear and a nonlinear function to combine the multiple outputs of FTA detectors, respectively. Moreover, since both the target and background spectra are used when calculating the autocorrelation matrix in FTA, the inversion of the autocorrelation matrix will suppress both the target and background information. In order to reduce the influence of multiple target signatures in the detection algorithms for multitemporal data, a measurement function between the target signatures and background signatures is proposed, and we modify the autocorrelation matrix of FTA.

The remainder of this article is organized as follows. Section II presents the background knowledge of CEM, the algorithms derived from CEM, and FTA. Section III introduces our proposed algorithms. Section IV describes the experiments and illustrates the performance of our proposed algorithms. Then, Section V displays the computational complexities and an extra attribute of our proposed methods. Finally, conclusions are exhibited in Section VI.

II. Background

In this section, we introduce some basic algorithms in the field of target detection in hyper/multispectral remote sensing images: CEM, the algorithms developed from CEM, and the basic knowledge of FTA.

A. CEM

As proposed by Harsanyi [37] in 1993, CEM is a single-target detection algorithm for single-temporal hyper/multispectral images. The idea of CEM, which is to attenuate and suppress other possible interference signals while keeping the output of specific signals constant, comes from the linear constrained minimum variance beamforming in the field of digital signal processing.

For a hyper/multispectral image $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ (N is the number of pixels and $\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{iL}]^{\mathrm{T}}$, L is the number of bands, $i = 1, \ldots, N$) and a given target signature $\mathbf{d} = [d_1, d_2, \ldots, d_L]^{\mathrm{T}}$, CEM algorithm implements an FIR filter $\mathbf{w} = [w_1, w_2, \ldots, w_L]^{\mathrm{T}}$ to suppress the background energy, while the target output remains constant:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

$\mathbf{d}^{\mathrm{T}}\mathbf{w} = \sum_{j=1}^{L} d_j w_j = 1$, which yields

$$\begin{cases} \min_{\mathbf{w}} \dfrac{1}{N}\left(\sum_{i=1}^{N} {y_i}^2\right) = \min_{\mathbf{w}} \mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w} \\ \text{s.t. } \mathbf{d}^{\mathrm{T}}\mathbf{w} = 1 \end{cases} \tag{1}$$

where $y_i = \mathbf{w}^{\mathrm{T}}\mathbf{x}_i$ is the output of CEM detector. $\mathbf{R} = 1/N \sum_{i=1}^{N} \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}$ is the autocorrelation matrix of the input image $\mathbf{X}$. The CEM detector $\mathbf{w}_{\mathrm{CEM}}$ can be computed from (1) as

$$\mathbf{w}_{\mathrm{CEM}} = \frac{\mathbf{R}^{-1}\mathbf{d}}{\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}}. \tag{2}$$

B. Single-Temporal Multitarget Detection Algorithms Derived From CEM

In order to apply CEM on multitarget detection problems, MTCEM [54] is implemented through adding more target signatures to the constraints of CEM. We assume that $\mathbf{D} = \left[\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_q\right]$ is a matrix containing q target signatures ($q \geq 2$), where $\mathbf{d}_i = [d_{i1}, d_{i2}, \ldots, d_{iL}]^{\mathrm{T}}$, $i = 1, 2, \ldots, q$. We substitute the target constraint in (1) for $\mathbf{D}^{\mathrm{T}}\mathbf{w} = \mathbf{1}$, where $\mathbf{1} = [1, 1, \ldots, 1]^{\mathrm{T}}$ (a q-dimensional column vector). Then, the model of MTCEM becomes

$$\begin{cases} \min_{\mathbf{w}} \mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w} \\ \text{s.t. } \mathbf{D}^{\mathrm{T}}\mathbf{w} = \mathbf{1}. \end{cases} \tag{3}$$

The MTCEM detector is expressed as

$$\mathbf{w}_{\mathrm{MTCEM}} = \mathbf{R}^{-1}\mathbf{D}\left(\mathbf{D}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{D}\right)^{-1}\mathbf{1}. \tag{4}$$

For reducing the effects of the desired targets in the autocorrelation matrix $\mathbf{R}$ of MTCEM, Geng and Zhao [60] modified the autocorrelation matrix, which can be expressed as

$$\widetilde{\mathbf{R}} = \mathbf{R} - \frac{1}{N}\sum_{i=1}^{q}\sum_{j=1}^{N}\left(\alpha_{ij}\right)^2 \mathbf{d}_i\mathbf{d}_i^{\mathrm{T}} \tag{5}$$

where α_{ij} is the abundance fraction of the ith ($i = 1, 2, \ldots, q$) target in the image and $\alpha_{ij} = \left(\mathbf{d}_i^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{r}_j\right)/\left(\mathbf{d}_i^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}_i\right)$. The modified algorithm is named revised-MTCEM (RMTCEM), and the RMTCEM detector is expressed as

$$\mathbf{w}_{\mathrm{RMTCEM}} = \widetilde{\mathbf{R}}^{-1}\mathbf{D}\left(\mathbf{D}^{\mathrm{T}}\widetilde{\mathbf{R}}^{-1}\mathbf{D}\right)^{-1}\mathbf{1}. \tag{6}$$

On the other hand, let $\left\{\mathrm{CEM}_j(\mathbf{r})\right\}_{j=1}^{q}$ be q ($q \geq 1$) target abundance fractions generated by q CEM detectors. SCEM [54] is defined as the sum of q outputs of CEM as

$$\mathrm{SCEM}(\mathbf{r}) = \sum_{j=1}^{q}\mathrm{CEM}_j(\mathbf{r}). \tag{7}$$

WTACEM [54] selects the maximum value at each pixel from q outputs of CEM as

$$\mathrm{WTACEM}(\mathbf{r}) = \max\left\{\mathrm{CEM}_j(\mathbf{r})\right\}_{j=1}^{q}. \tag{8}$$

C. FTA

Traditional target detection methods for multispectral data, such as CEM, MTCEM, SCEM, and WTACEM, are originally constructed for single-temporal multispectral data, which cannot fully make use of the time-series information. Since the spectra of targets change over time and, sometimes, the difference between the spectra of target and background is not so obvious in a given time phase, it is of great importance to use multitemporal data in target detection. Thus, FTA [59] is proposed to detect targets using multitemporal data.

Suppose that $\mathbf{X}^{(t)} = \{\mathbf{r}_1^{(t)}, \mathbf{r}_2^{(t)}, \ldots, \mathbf{r}_N^{(t)}\}$ represents the remote sensing data at the time phase t (there are M time phases, $t = 1, 2, \ldots, M$, and N is the total number of pixels) and the corresponding number of bands are $L_1, L_2, \ldots, L_M$, which can be different at different time phase t. Then, for the ith pixel in the data, it is composed of M vectors and denoted by $\{\mathbf{r}_i^{(1)}, \mathbf{r}_i^{(2)}, \ldots, \mathbf{r}_i^{(M)}\}$. Thus, a given target $\mathbf{d}$ also consists of M vectors, which can be denoted by $\{\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \ldots, \mathbf{d}^{(M)}\}$.

With an Mth-order tensor filter $\mathscr{W}$ that has a dimension of $L_1 \times L_2 \times \cdots \times L_M$, FTA minimizes the filter output energy, while the output of target is equal to one ("output-to-one" constraint). FTA can be formulated as

$$\begin{cases} \min_{\mathscr{W}} \dfrac{1}{N}\displaystyle\sum_{i}\left(\mathscr{W} \times_1 \mathbf{r}_i^{(1)} \times_2 \mathbf{r}_i^{(2)} \cdots \times_M \mathbf{r}_i^{(M)}\right)^2 \\ \text{s.t. } \mathscr{W} \times_1 \mathbf{d}^{(1)} \times_2 \mathbf{d}^{(2)} \cdots \times_M \mathbf{d}^{(M)} = 1 \end{cases} \tag{9}$$

where the operator $\times_j$ ($j = 1, 2, \ldots, M$) denotes the j-mode product operator.

As is known from the fundamental theory of matrix [61], tensor product can be calculated by the Kronecker product as

$$\begin{aligned} &\mathscr{W} \times_1 \mathbf{r}_i^{(1)} \times_2 \mathbf{r}_i^{(2)} \cdots \times_M \mathbf{r}_i^{(M)} \\ &\quad = \left(\mathbf{r}_i^{(M)} \otimes \cdots \otimes \mathbf{r}_2^{(2)} \otimes \mathbf{r}_i^{(1)}\right)^{\mathrm{T}} \mathrm{vec}(\mathscr{W}) \end{aligned} \tag{10}$$

$$\begin{aligned} &\mathscr{W} \times_1 \mathbf{d}^{(1)} \times_2 \mathbf{d}^{(2)} \cdots \times_M \mathbf{d}^{(M)} \\ &\quad = \left(\mathbf{d}^{(M)} \otimes \cdots \otimes \mathbf{d}^{(2)} \otimes \mathbf{d}^{(1)}\right)^{\mathrm{T}} \mathrm{vec}(\mathscr{W}) \end{aligned} \tag{11}$$

where $\otimes$ represents the Kronecker product operator, and vec() is defined as the vectorization operator, which can transform an $(L_1 \times L_2 \times \cdots \times L_k)$ tensor (k is the order of the tensor) into an $(L_1 \times L_2 \times \cdots \times L_k)$-dimensional vector.

Thus, let

$$\hat{\mathbf{w}} = \mathrm{vec}(\mathscr{W}) \tag{12}$$

$$\hat{\mathbf{r}}_i = \mathbf{r}_i^{(M)} \otimes \cdots \otimes \mathbf{r}_i^{(2)} \otimes \mathbf{r}_i^{(1)} \tag{13}$$

$$\hat{\mathbf{R}} = \left(\sum_{i=1}^{N}\hat{\mathbf{r}}_i\hat{\mathbf{r}}_i^{\mathrm{T}}\right)/N \tag{14}$$

$$\hat{\mathbf{d}} = \mathbf{d}^{(M)} \otimes \cdots \otimes \mathbf{d}^{(2)} \otimes \mathbf{d}^{(1)}. \tag{15}$$

Then, (9) is transformed into

$$\begin{cases} \min_{\hat{\mathbf{w}}} \hat{\mathbf{w}}^{\mathrm{T}}\hat{\mathbf{R}}\hat{\mathbf{w}} \\ \text{s.t. } \hat{\mathbf{d}}^{\mathrm{T}}\hat{\mathbf{w}} = 1. \end{cases} \tag{16}$$

The vector form of FTA operator is

$$\hat{\mathbf{w}}_{\mathrm{FTA}} = \frac{\hat{\mathbf{R}}^{-1}\hat{\mathbf{d}}}{\hat{\mathbf{d}}^{\mathrm{T}}\hat{\mathbf{R}}^{-1}\hat{\mathbf{d}}}. \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

The tensor form of FTA operator can be written as

$$\mathscr{W}_{\mathrm{FTA}} = \mathrm{ten}(\hat{\mathbf{w}}_{\mathrm{FTA}}) \tag{18}$$

where ten() is the reverse operator of vec() to reform an $(L_1 \times L_2 \times \cdots \times L_k)$-dimensional vector into an $(L_1 \times L_2 \times \cdots \times L_k)$ tensor [61].

III. Proposed Methods

Although FTA has shown some success in multitemporal multispectral images, it is very sensitive to the selection of the target signature. As a result, FTA can only detect the target whose spectra have little variability. Yet, due to the phenomenon that the same ground object has different spectra, we often have to select multiple spectra to detect the whole desired ground object. In addition, in practical applications, there is a need to extract multiple targets simultaneously. Therefore, in this section, we introduce several algorithms for multitarget detection in multitemporal data.

A. Multiple-Target FTA (MTFTA)

We assume that, in the multitemporal data $\mathbf{X}^{(t)} = \{\mathbf{r}_1^{(t)}, \mathbf{r}_2^{(t)}, \ldots, \mathbf{r}_N^{(t)}\}$, there are q targets of interest ($q \geq 2$), and each target signature $\mathbf{d}_i$ ($i = 1, 2, \ldots, q$) consists of M vectors $\{\mathbf{d}_i^{(1)}, \mathbf{d}_i^{(2)}, \ldots, \mathbf{d}_i^{(M)}\}$. Then, by expanding the "output-to-one" constraint in FTA in (9), the model of MTFTA becomes

$$\begin{cases} \min_{\mathscr{W}} \dfrac{1}{N}\left(\displaystyle\sum_i \mathscr{W} \times_1 \mathbf{r}_i^{(1)} \times_2 \mathbf{r}_i^{(2)} \cdots \times_M \mathbf{r}_i^{(M)}\right)^2 \\ \text{s.t.} \mathscr{W} \times_1 \mathbf{d}_1^{(1)} \times_2 \mathbf{d}_1^{(2)} \cdots \times_M \mathbf{d}_1^{(M)} = 1 \\ \qquad \mathscr{W} \times_1 \mathbf{d}_2^{(1)} \times_2 \mathbf{d}_2^{(2)} \cdots \times_M \mathbf{d}_2^{(M)} = 1 \\ \qquad \vdots \\ \qquad \mathscr{W} \times_1 \mathbf{d}_q^{(1)} \times_2 \mathbf{d}_q^{(2)} \cdots \times_M \mathbf{d}_q^{(M)} = 1 \end{cases} \tag{19}$$

where the operator $\times_j (j = 1, 2, \ldots, M)$ denotes the j-mode product operator.

Let

$$\hat{\mathbf{d}}_i = \mathbf{d}_i^{(M)} \otimes \cdots \otimes \mathbf{d}_i^{(2)} \otimes \mathbf{d}_i^{(1)} \tag{20}$$

and calculate the tensor products in (19) with the Kronecker products displayed in (10) and (11), and the model of MTFTA is simplified as

$$\begin{cases} \min_{\hat{\mathbf{w}}} \hat{\mathbf{w}}^{\mathrm{T}} \hat{\mathbf{R}} \hat{\mathbf{w}} \\ \text{s.t. } \hat{\mathbf{d}}_1^{\mathrm{T}} \hat{\mathbf{w}} = 1 \\ \qquad \hat{\mathbf{d}}_2^{\mathrm{T}} \hat{\mathbf{w}} = 1 \\ \qquad \vdots \\ \qquad \hat{\mathbf{d}}_q^{\mathrm{T}} \hat{\mathbf{w}} = 1. \end{cases} \tag{21}$$

We denote

$$\hat{\mathbf{D}} = \left[\hat{\mathbf{d}}_1, \hat{\mathbf{d}}_2, \ldots, \hat{\mathbf{d}}_q\right] \tag{22}$$

and (21) can be further simplified as

$$\begin{cases} \min_{\hat{\mathbf{w}}} \hat{\mathbf{w}}^{\mathrm{T}} \hat{\mathbf{R}} \hat{\mathbf{w}} \\ \text{s.t. } \hat{\mathbf{D}}^{\mathrm{T}} \hat{\mathbf{w}} = \mathbf{1} \end{cases} \tag{23}$$

where the vector $\mathbf{1}$ is defined in Section II-B.

Applying the method of the Lagrange multiplier to (23), we get the Lagrangian expression as

$$L(\hat{\mathbf{w}}, \boldsymbol{\lambda}) = \frac{1}{2} \hat{\mathbf{w}}^{\mathrm{T}} \hat{\mathbf{R}} \hat{\mathbf{w}} + \boldsymbol{\lambda}^{\mathrm{T}} \left(\mathbf{1} - \hat{\mathbf{D}}^{\mathrm{T}} \hat{\mathbf{w}}\right) \tag{24}$$

where $\boldsymbol{\lambda} = \left[\lambda_1, \lambda_2, \ldots, \lambda_q\right]^{\mathrm{T}}$. The partial derivative of (24) with respect to $\hat{\mathbf{w}}$ is

$$\frac{\partial L(\hat{\mathbf{w}}, \boldsymbol{\lambda})}{\partial \hat{\mathbf{w}}} = \hat{\mathbf{R}} \hat{\mathbf{w}} - \hat{\mathbf{D}} \boldsymbol{\lambda}. \tag{25}$$

Let $(\partial L(\hat{\mathbf{w}}, \boldsymbol{\lambda}))/(\partial \hat{\mathbf{w}}) = 0$, and $\hat{\mathbf{w}}$ can be expressed as

$$\hat{\mathbf{w}} = \hat{\mathbf{R}}^{-1} \hat{\mathbf{D}} \boldsymbol{\lambda}. \tag{26}$$

Substitute (26) into (23), and we can have

$$\boldsymbol{\lambda} = \left(\hat{\mathbf{D}}^{\mathrm{T}} \hat{\mathbf{R}}^{-1} \hat{\mathbf{D}}\right)^{-1} \mathbf{1}. \tag{27}$$

Then, we substitute (27) into (26), the vector form of the MTFTA operator can be expressed as follows:

$$\hat{\mathbf{w}}_{\mathrm{MTFTA}} = \hat{\mathbf{R}}^{-1} \hat{\mathbf{D}} \left(\hat{\mathbf{D}}^{\mathrm{T}} \hat{\mathbf{R}}^{-1} \hat{\mathbf{D}}\right)^{-1} \mathbf{1}. \tag{28}$$

The output of MTFTA for a given pixel $\hat{\mathbf{r}}_i$ is computed as

$$y_i = \hat{\mathbf{w}}_{\mathrm{MTFTA}}^{\mathrm{T}} \hat{\mathbf{r}}_i = \left(\hat{\mathbf{R}}^{-1} \hat{\mathbf{D}} \left(\hat{\mathbf{D}}^{\mathrm{T}} \hat{\mathbf{R}}^{-1} \hat{\mathbf{D}}\right)^{-1} \mathbf{1}\right)^{\mathrm{T}} \hat{\mathbf{r}}_i. \tag{29}$$

The pseudo code for MTFTA is presented as follows.

Algorithm 1 MTFTA

Input: multitemporal multispectral data $\mathbf{X}^{(t)} = \{\mathbf{r}_1^{(t)}, \mathbf{r}_2^{(t)}, \ldots, \mathbf{r}_N^{(t)}\}$ and target signatures $\mathbf{d}_i$ ($i = 1, 2, \ldots, q$).
Output: detection result $\mathbf{Y} = \{y_1, y_2, \ldots, y_N\}$.

1: Based on (20) and (22), construct the target matrix $\hat{\mathbf{D}}$.
2: According to (13) and (14), calculate the autocorrelation matrix $\hat{\mathbf{R}}$.
3: Calculate the MTFTA projection operator $\hat{\mathbf{w}}$ based on (28).
4: Compute the output y_i by applying $\hat{\mathbf{w}}$ to pixel $\hat{\mathbf{r}}_i$ according to (29).

B. Orthogonal Subspace Projection Weighted MTFTA (OWMTFTA)

In the algorithms, such as CEM and MTCEM, the target pixels and background pixels contribute to the autocorrelation matrix equally, which brings about the consequence that the inversion of the autocorrelation matrix suppresses both the background and target. To fix this problem, Geng and Zhao [60] proposed a weighted correlation matrix $\mathbf{R}^*$ that eliminates the effects of target pixels in the construction of the autocorrelation matrix, and $\mathbf{R}^*$ is defined as

$$\mathbf{R}^* = c \sum_{i=1}^{N} f(g(\mathbf{r}_i, \mathbf{d})) \mathbf{r}_i \mathbf{r}_i^{\mathrm{T}} \tag{30}$$

where c is a constant. $f(x)$ is a monotonically increasing function for $x > 0$ and $f(x) = 0$ when $x = 0$. $g(\mathbf{u}_1, \mathbf{u}_2)$ is a function for measuring the spectral similarity between $\mathbf{u}_1$ and $\mathbf{u}_2$, which increases as the similarity of $\mathbf{u}_1$ and $\mathbf{u}_2$ decreases. For example, considering $\mathbf{u}_2$ is the target vector,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

we have $f(g(\mathbf{u}_1, \mathbf{u}_2)) = 0$ when $\mathbf{u}_1$ is completely the same as $\mathbf{u}_2$, which keeps the spectral vector $\mathbf{u}_1$ from the calculation of the autocorrelation matrix $\mathbf{R}^*$. When the vector $\mathbf{u}_1$ comes from a background pixel, we have $f(g(\mathbf{u}_1, \mathbf{u}_2)) > 0$ so that the spectral vector $\mathbf{u}_1$ would contribute to $\mathbf{R}^*$. Ji *et al.* [62] also proposed an orthogonal subspace projection-weighted (OW) autocorrelation matrix in detecting large area of water using CEM.

In MTFTA, the function of $\hat{\mathbf{R}}^{-1}$ is to suppress the background information. It is optimal that the autocorrelation matrix $\hat{\mathbf{R}}$ is only composed of the background pixels. However, in practical applications, it is difficult to distinguish the target pixels from the background ones in constructing $\hat{\mathbf{R}}$. When there are few target pixels in the image, the influence of target pixels on $\hat{\mathbf{R}}$ can be ignored. However, the impact cannot be neglected when there are many target pixels.

With $\hat{\mathbf{D}}$ representing a matrix of multiple targets as in (23) and $\hat{\mathbf{r}}$ being a spectral vector as in Section II-C, we use the OSP operator $\mathbf{C} = \mathbf{I} - \hat{\mathbf{D}}\hat{\mathbf{D}}^{+}$ to separate the target from background, where $\hat{\mathbf{D}}^{+}$ is the pseudo inversion of $\hat{\mathbf{D}}$. The OSP operator projects a given vector $\hat{\mathbf{r}}$ into the orthogonal complement space of the column vectors in matrix $\hat{\mathbf{D}}$; thus, $\hat{\mathbf{r}}^{\mathrm{T}}\mathbf{C}\hat{\mathbf{r}}$ represents the length of vector $\hat{\mathbf{r}}$ in the orthogonal complement space of the column vectors in $\hat{\mathbf{D}}$, which we define as $g(\hat{\mathbf{r}}, \hat{\mathbf{D}})$. Thus, with $g(\hat{\mathbf{r}}_i, \hat{\mathbf{D}}) = \hat{\mathbf{r}}_i^{\mathrm{T}}\mathbf{C}\hat{\mathbf{r}}_i = \hat{\mathbf{r}}_i^{\mathrm{T}}(\mathbf{I} - \hat{\mathbf{D}}\hat{\mathbf{D}}^{+})\hat{\mathbf{r}}_i$, $c = 1/N$ and $f(x) = x$, the weighted autocorrelation matrix of MTFTA is modified as

$$\hat{\mathbf{R}}^{*} = \frac{1}{N}\sum_{i=1}^{N}\hat{\mathbf{r}}_i^{\mathrm{T}}(\mathbf{I} - \hat{\mathbf{D}}\hat{\mathbf{D}}^{+})\hat{\mathbf{r}}_i\hat{\mathbf{r}}_i\hat{\mathbf{r}}_i^{\mathrm{T}}. \tag{31}$$

In this article, we call $\hat{\mathbf{R}}^{*}$ as the Orthogonal subspace projection Weighted (OW) autocorrelation matrix and this modified version of MTFTA algorithm as OWMTFTA, of which the detector is expressed as

$$\hat{\mathbf{w}}_{\mathrm{OWMTFTA}} = \hat{\mathbf{R}}^{*-1}\hat{\mathbf{D}}\left(\hat{\mathbf{D}}^{\mathrm{T}}\hat{\mathbf{R}}^{*-1}\hat{\mathbf{D}}\right)^{-1}\mathbf{1}. \tag{32}$$

From the derivation of OWMTFTA, we can see that it has the same computation process as MTFTA, except for the modification of the autocorrelation matrix. Hence, to perform OWMTFTA, we only need to replace the calculation of $\hat{\mathbf{R}}$ with $\hat{\mathbf{R}}^{*}$ in step 2 in Algorithm 1.

C. Sum FTA (SFTA) and Winner-Take-All FTA (WTAFTA)

With the same construction patterns of SCEM and WTACEM for the detection of multiple targets using a single-target method, we here introduce SFTA and WTAFTA.

Let $\mathbf{r}$ denote an arbitrary pixel from the multitemporal data and $\{\mathrm{FTA}_j(\mathbf{r})\}_{j=1}^{q}$ represent the q target abundance fractions computed by q ($q \geq 1$) FTA detectors. Then, the SFTA detector simply sums up q FTA outputs, which can be described as

$$\mathrm{SFTA}(\mathbf{r}) = \sum_{j=1}^{q}\mathrm{FTA}_j(\mathbf{r}). \tag{33}$$

WTAFTA selects the maximum value at each pixel from q FTA detectors. It is defined as

$$\mathrm{WTAFTA}(\mathbf{r}) = \max\{\mathrm{FTA}_j(\mathbf{r})\}_{j=1}^{q}. \tag{34}$$

Implementing SFTA/WTAFTA is to first perform FTA q times by q different target signature's $\mathbf{d}_i$'s and then calculate the output according to (33) and (34).

D. Remarks About the Proposed Methods

As is presented earlier, we have proposed four multitarget detection algorithms for multitemporal remote sensing data. Here, we will discuss the differences and advantages of those methods in detail.

Through the construction of MTFTA shown in (21), compared with FTA, MTFTA has the ability to detect multiple target signatures simultaneously because it adds the "output-to-one" constraints of all targets in the design of the tensor filter. From a computational point of view, the detection result of all targets is computed at once by MTFTA. Yet, it may not perform well when the target spectrum does not change much in different time phases.

Furthermore, OWMTFTA aims to reduce the influence of included target signatures in calculating the autocorrelation matrix of MTFTA. Theoretically, OWMTFTA can detect the targets well whether the targets are "small" or not, according to its construction of the autocorrelation matrix. Yet, in real cases, due to the incident radiance difference and growing status change of the ground objects, there exist some fluctuations in the reflectance of one ground object. Therefore, it is still difficult to completely remove the influence of target spectra in the autocorrelation matrix in real applications. From the aspect of computation, because OWMTFTA always needs to calculate the modified autocorrelation matrix first, it requires the most computational time and memory among the four algorithms, which will be explained in Section V.

From the implementation of SFTA, we know it combines the target detection results of FTA for each of the multiple targets with a linear function "sum". That is, it adds up the abundance fraction in each output of FTA. Hence, to some extent, we can regard SFTA as decomposing a complex task of detecting multiple targets into a combination of detecting several single targets. From this perspective, it can effectively alleviate the impact of "the same ground object with different spectra." Especially, when the spectrum of the target changes little spatially, it can extract all the target pixels. However, SFTA tends to overestimate the abundance fraction for each target.

Finally, WTAFTA uses a nonlinear function "maximum" to combine the results of FTA for each of the multiple targets, which is also known as the winner-take-all rule to determine the gray-scale value of each pixel [54]. It can also solve the problem of "the same ground object having different spectra" to some extent. Moreover, compared with SFTA, WTAFTA can avoid the overestimation of the abundance fraction for multiple targets since the value of a pixel is only determined by the largest value. Because SFTA and WTAFTA both need the detection results of all the single target, they share the same computational time, which will be demonstrated in Section V.

More comparisons about the computation details of the four algorithms can be found in Section V.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BAND DESIGNATIONS FOR LANDSAT 7 AND LANDSAT 8

	Coastal blue	Blue	Green	Red	Near infrared (NIR)	Short-wave infrared 1 (SWIR1)	Short-wave infrared 2 (SWIR2)
Landsat 7	–	1	2	3	4	5	6
Landsat 8	1	2	3	4	5	6	7

TABLE II
CONFUSION MATRIX FOR THE TARGET DETECTION OUTPUT

	True Target	True Non-Target
Hypothesized Target	True Positive (TP)	False Positive (FP)
Hypothesized Non-Target	False Negative (FN)	True Negative (TN)

IV. EXPERIMENTS AND RESULTS

In this section, three sets of data are used to evaluate the effectiveness of our proposed algorithms. The data include one set of simulation data and two sets of Landsat 7/8 data, named: 1) the farmland data and 2) the airport and roof data. The band information of Landsat 7 and Landsat 8 images is shown in Table I. We compare our proposed methods (MTFTA, OWMTFTA, SFTA, and WTAFTA) with the single-temporal single-target methods (CE, SWCEM, and E-CEM) and the single-temporal multitarget detection methods mentioned in Section II (MTCEM, RMTCEM, SCEM, and WTACEM) on the three data sets. The codes of CE and E-CEM are provided by their authors, while the other algorithms are implemented by ourselves. Since CE, SWCEM, and E-CEM need only one target spectrum, we average all the target spectra as the desired target spectrum. When implementing those seven single-temporal methods, we combine the data of all time phases into one single-temporal data.

With respect to the quantitative accuracy evaluation, the receiver operating characteristic (ROC) curve, area under the ROC curve (AUC), overall accuracy (OA), F-score, and Cohen's kappa coefficient [63] are used. The ROC curve and the corresponding AUC value are calculated from the output of each method directly, while OA, F-score, and Cohen's kappa coefficient are computed from the binary classification of the detection results. It should be noted here that, since the targets in the experiments of real Landsat 7/8 multispectral data are all "small" targets, the size of reference samples for the background (non-target) is much larger than that for targets. For a fair comparison, we randomly select the background pixels of twice the number of target pixels [64]. The average OA, F-score, and kappa coefficient of 20 runs are computed in the experiments of the real Landsat 7/8 multispectral data. To obtain the binary classification, we use the thresholding method, in which the optimal threshold is determined from the ROC curve by Youden's index [65]–[67]. According to the confusion matrix (see Table II) from the binary classification image, OA can be computed as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{35}$$

F-score is defined as

$$\text{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{36}$$

$$\text{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{37}$$

$$\text{F-score} = \left(1 + \beta^2\right) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}} \tag{38}$$

where β is the weight of precision in calculating F-score. By setting $\beta = 1$, we make the precision and recall have the equal weight in F-score.

The kappa coefficient is computed as

$$\mathrm{P_e} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{FP} + \mathrm{TN})(\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN})^2} \tag{39}$$

$$\text{kappa coefficient} = \frac{\mathrm{OA} - \mathrm{P_e}}{1 - \mathrm{P_e}}. \tag{40}$$

It should be clear that ROC and AUC are calculated from a list of thresholds in the detection result, while OA, F-score, and Cohen's kappa coefficient are determined from the classification result using only one threshold determined by Youden's index. Thus, these two kinds of accuracy metrics are not totally equivalent.

A. Simulation Data

We synthesize a set of multispectral data containing three time phases (time phases 1–3), each of which has a size of $200 \times 200 \times 7$. Then, we set up four targets, and each target contains 70×70 pixels. For explanatory convenience, we mark the four targets as targets 1–4, respectively [see Fig. 1(a)]. As we know, different ground objects may have similar spectra in a given time phase, while a single object may reveal different spectra in some time phases. To simulate this phenomenon, we make target 1 and 3 have similar spectra as the background in all time phases, especially target 3 [see Fig. 1(b)–(e)]. Then, we add Gaussian white noise with a zero mean and SNR = 20 dB to the whole data.

From the distribution of simulation data in Fig. 1(f), we can see that the pixels of target 1 are very close to the background pixels, while the pixels of target 3 are mixed with the background pixels. Meanwhile, because the spectra of targets 2 and 4 are different from that of background in time phase 1 and 3, their pixels are far away from the background.

The detection results of those algorithms are presented in Fig. 2. Visually, although the pixels are mixed with noise, our proposed FTA-based methods can suppress the background better than those single-temporal single-target algorithms

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE III

AUC, OA, F-SCORE, AND KAPPA COEFFICIENT OF THE DETECTION RESULTS OF DIFFERENT ALGORITHMS FOR THE SIMULATION DATA. BOLD NUMBERS INDICATE THE HIGHEST VALUE IN EACH ROW

	CE	SWCEM	E-CEM	MTCEM	RMTCEM	SCEM	WTACEM	MTFTA	OWMTFTA	SFTA	WTAFTA
AUC	0.9969	0.9896	0.9838	0.9597	0.9597	0.9686	0.9844	**0.9990**	0.9980	0.9989	**0.9990**
OA	0.9747	0.9496	0.9366	0.8909	0.8909	0.9057	0.9441	**0.9889**	0.9838	0.9887	0.9860
F-score	0.9741	0.9484	0.9353	0.8886	0.8886	0.9035	0.9429	**0.9886**	0.9834	0.9885	0.9857
Kappa coefficient	0.9494	0.8991	0.8732	0.7817	0.7817	0.8114	0.8882	**0.9777**	0.9676	0.9775	0.9720

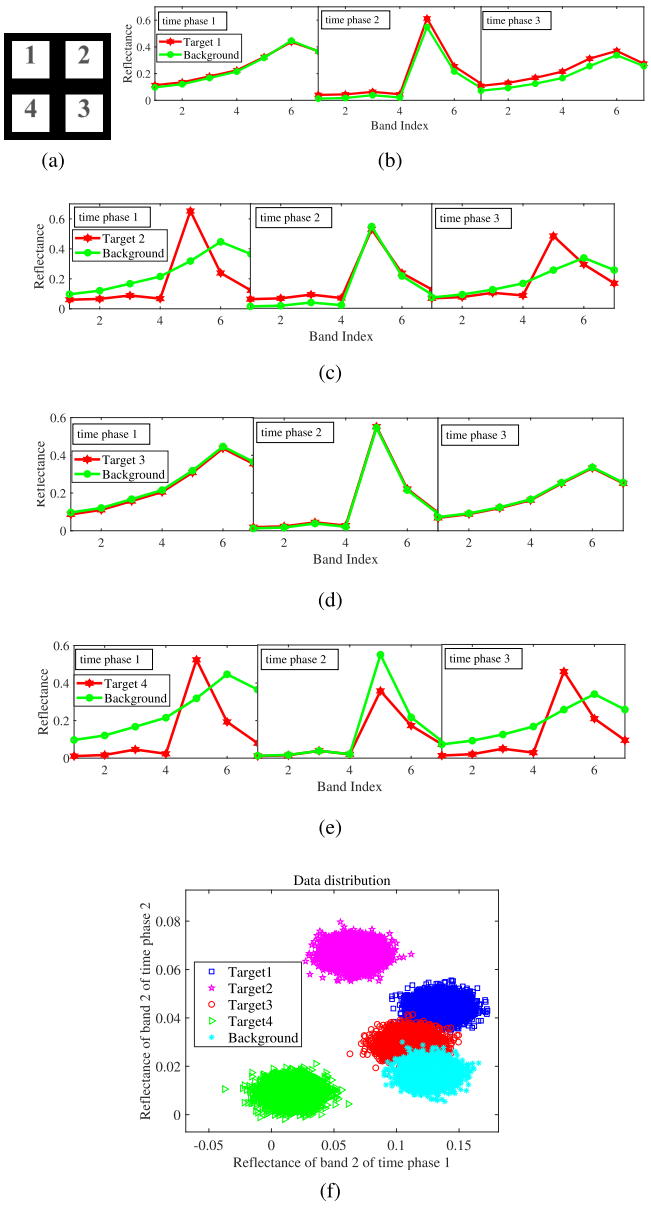


Fig. 1. Simulation data. (a) Ground-truth image. (b) Spectral signatures of target 1. (c) Spectral signatures of target 2. (d) Spectral signatures of target 3. (e) Spectral signatures of target 4. (f) Distribution of simulation data.

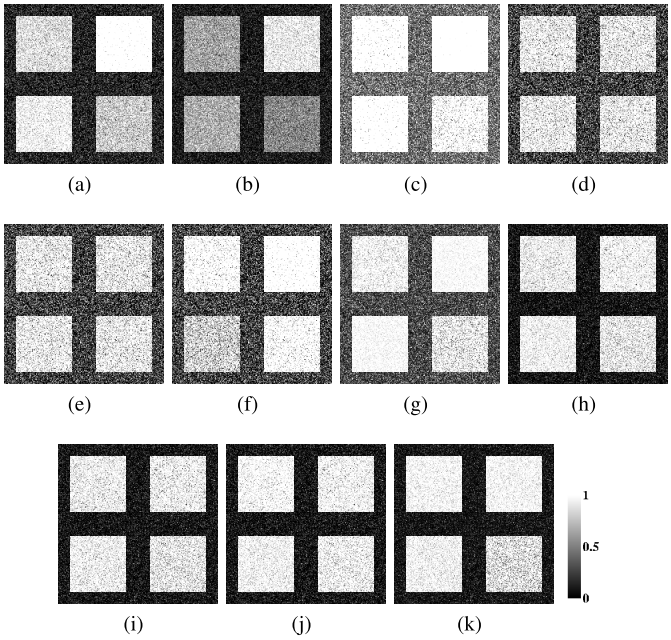


Fig. 2. Target detection results for the simulation data. (a) CE. (b) SWCEM. (c) E-CEM. (d) MTCEM. (e) RMTCEM. (f) SCEM. (g) WTACEM. (h) MTFTA. (i) OWMTFTA. (j) SFTA. (k) WTAFTA.

(CE, SWCEM, and E-CEM) and single-temporal multitarget algorithms (MTCEM, RMTCEM, SCEM, and WTACEM). From Fig. 2(a) and (c), we can see that the single-temporal single-target algorithms are not able to detect all the targets at one time; it is because they only use one spectral signature to represent all the targets. Because of the spectral similarity of some targets and background in one or two time phases, the contrast between the targets and background in the results of MTCEM, RMTCEM, SCEM, and WTACEM are not very obvious [see Fig. 2(d)–(g)]. However, our FTA-based methods can achieve better target detection [see Fig. 2(h)–(k)] in this situation, which indicates the effectiveness of our methods over the single-temporal algorithms.

In order to quantitatively compare the accuracy of those detection algorithms, their ROC curves are plotted in Fig. 3, and the corresponding AUC values are tabulated in Table III. It is seen that the multitemporal multitarget detection algorithms derived from FTA have better ROC curves than those single-temporal algorithms. Similarly, the four FTA-based algorithms have higher AUC values than the others. Among them, MTFTA and WTAFTA perform the best. The reason for MTFTA's excellent performance lies in that it is designed to simultaneously detect multiple targets by setting the "output-to-one" constraints for all targets. Hence, it is able to detect all four targets with high accuracy. As for WTAFTA, it picks the maximum value from the four FTA results for each pixel,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

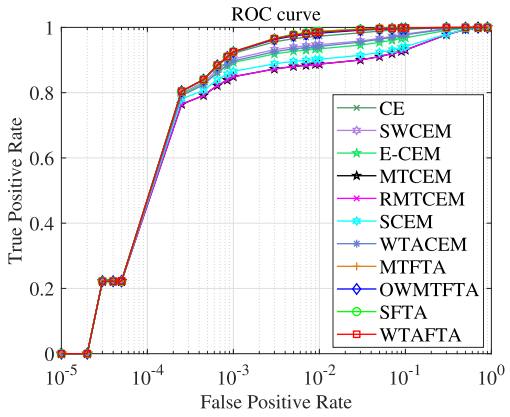


Fig. 3. ROC curves of the detection results of CE, SWCEM, E-CEM, MTCEM, RMTCEM, SCEM, WTACEM, MTFTA, OWMTFTA, SFTA, and WTAFTA for the simulation data.

so it is able to alleviate the influence of noise to some extent. The AUC values of the other two FTA-based methods are slightly lower. It can be attributed to the fact that they are more sensitive to noise. SFTA accumulates the effect of the noise of the four FTA outputs. For OWMTFTA, since it uses the target spectra to generate the weight values used in $\hat{\mathbf{R}}^*$ [see (31)], the spectral inconsistency of the targets due to the added noise is more likely to cause a more serious impact on the detection result of OWMTFTA than that of MTFTA. Moreover, Table III also shows the accuracy metrics (OA, F-score, and kappa coefficient) of different algorithms based on the binary classification, from which we can observe that our proposed algorithms perform better than the single-temporal single-target algorithms and the single-temporal multitarget algorithms.

B. Farmland Data

The farmland data include two Landsat 8 images (each image has a size of 115 × 208 × 7) in different seasons of 2014 with day of year (DOY) equal to 244 (summer) and 308 (winter) [see Fig. 4(a) and (b)]. The study area is located in Nebraska, USA, and mainly covered by six kinds of ground objects: soybeans, corn, grassland, peas, winter wheat, and alfalfa. The ground-truth map [see Fig. 4(c)] is determined from the cropland data layer (CDL) in 2014 from the United States Department of Agriculture website (https://nassgeodata.gmu.edu/CropScape/).

The winter wheat field in brown (marked by the green arrow) and the alfalfa field in magenta (marked by the red arrow) in Fig. 4(c) are selected as the targets of interest. From Fig. 4(d), it is easily seen that the winter wheat field has shown the spectral characteristics of bare land in both time phases. From Fig. 4(e), we can observe that the alfalfa field has constantly been vegetation. Therefore, the two targets have different growing patterns through the two time phases.

The target detection results for the farmland data are shown in Fig. 5. From the visual point of view, our proposed algorithms can suppress the non-target signals much better than those single-temporal algorithms. In Fig. 5(a) and (b), CE and SWCEM cannot detect the two targets effectively.

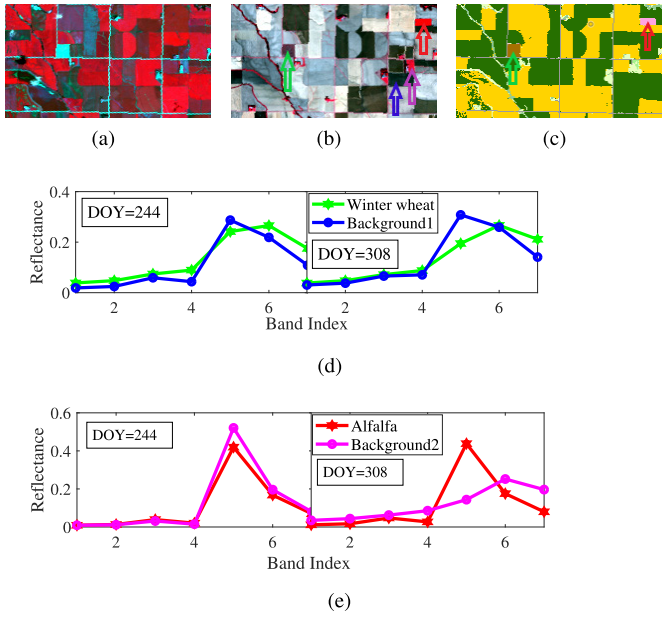


Fig. 4. Farmland data (Landsat 8, path=029, and row=031) in 2014. (a) False-color image (R: band 5; G: band 4; and B: band 3) with DOY = 244. (b) False-color image with DOY = 308 (the winter wheat is marked by the green arrow; the alfalfa is marked by the red arrow; the background marked by the blue arrow is called background 1; and the background marked by the purple arrow is called background 2). (c) Ground-truth image derived from 2014 CDL data (dark green: soybeans; yellow: corn; lemon chiffon: grassland; light green: peas; brown: winter wheat (also marked by the green arrow); and magenta: alfalfa [also marked by the red arrow)]. (d) Spectral signatures of the winter wheat and background 1. (e) Spectral signatures of the alfalfa and background 2.

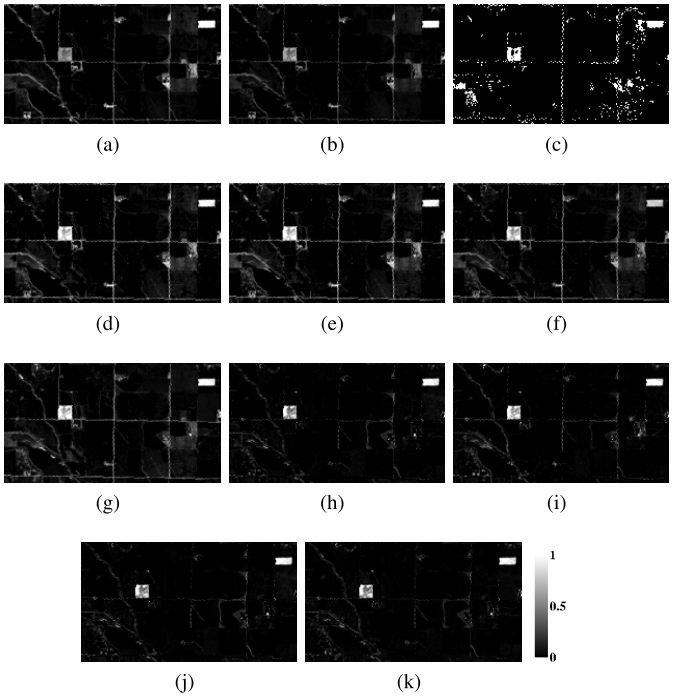


Fig. 5. Target detection results for the farmland data. (a) CE. (b) SWCEM. (c) E-CEM. (d) MTCEM. (e) RMTCEM. (f) SCEM. (g) WTACEM. (h) MTFTA. (i) OWMTFTA. (j) SFTA. (k) WTAFTA.

In the detection result of E-CEM [see Fig. 5(c)], there are still many misclassified background pixels. In Fig. 5(d)–(g), using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
AUC, OA, F-SCORE, AND KAPPA COEFFICIENT OF DIFFERENT ALGORITHMS FOR THE FARMLAND DATA. BOLD NUMBERS INDICATE THE HIGHEST VALUE IN EACH ROW

	CE	SWCEM	E-CEM	MTCEM	RMTCEM	SCEM	WTACEM	MTFTA	OWMTFTA	SFTA	WTAFTA
AUC	0.9888	0.9908	0.8979	0.9893	0.9893	0.9883	0.9917	0.9938	0.9955	0.9938	**0.9957**
OA	0.9588	0.9529	0.9552	0.9609	0.9609	0.9573	0.9640	0.9787	0.9704	0.9765	**0.9810**
F-score	0.9394	0.9311	0.9344	0.9410	0.9410	0.9363	0.9460	0.9680	0.9561	0.9648	**0.9717**
Kappa coefficient	0.9012	0.8960	0.8927	0.9050	0.9050	0.9207	0.9234	0.9578	0.9268	0.9370	**0.9657**

the data of both time phases combined, we can see that the single-temporal multitarget methods (MTCEM, RMTCEM, SCEM, and WTACEM) perform not well in suppressing the background. The reason lies in the fact that there exist background pixels that have similar spectral signatures as the target pixels in some time phases. For example, background 1 [marked by the blue arrow in Fig. 4(b)] and the winter wheat field have very similar spectrum in both time phases [see Fig. 4(d)]; background 2 [marked by the purple arrow in Fig. 4(b)] has similar spectrum as the alfalfa field in the first time phase [see Fig. 4(e)]. As a result, those background pixels cannot be well suppressed by MTCEM, RMTCEM, SCEM, and WTACEM. However, although the spectral signatures of the targets are similar to some background pixels in one or two time phases, our FTA-based methods can detect the targets better than those single-temporal methods [see Fig. 5(h)–(k)].

The ROC curves of the eleven different algorithms are plotted in Fig. 6, and the AUC values are listed in Table IV. There is an obvious gap between the ROC curves of our multitemporal multitarget algorithms and those of single-temporal algorithms. Moreover, OA, F-score, and kappa coefficient (see Table IV) also indicate that our FTA-based methods can always reach higher binary classification accuracy than the single-temporal methods. Among them, WTAFTA has the best performance by achieving the highest values for all the four accuracy metrics. The accuracy metrics of the other three FTA-based algorithms are similar but all lower than those of WTAFTA. The construction of WTAFTA is similar to SFTA, but the former one has a higher OA, F-score, and kappa coefficient. It infers that, for the farmland data, the nonlinear function "maximum" used in WTAFTA is more effective than the "sum" function in SFTA. OWMTFTA modifies the autocorrelation matrix in MTFTA in order to reduce the influence of targets spectra on the autocorrelation matrix. However, the two targets in the data are both "small" targets. As a result, the modification of the autocorrelation matrix in OWMTFTA shows little effect in improving the detection accuracy (AUC in OWMTFTA is just 0.0017 higher than that in MTFTA).

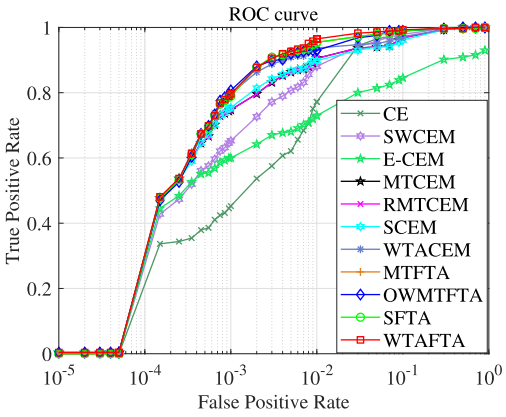


Fig. 6. ROC curves of the detection results of different algorithms for the farmland data. The x-axis is plotted in the logarithmic scale.

C. Airport and Roof Data

In this experiment, we use one Landsat 7 image (DOY=010 and year=2000) with a dimension of 400 × 400 × 6 and one Landsat 8 image (DOY=045 and year=2014) with a size of 400 × 400 × 7 [see Fig. 7(a) and (b)] to study the effectiveness of our proposed methods. A newly built airport and roof (marked by the green and red arrow in Fig. 7(b), respectively) are selected as targets. The study area is located in Mudug, Somalia. The ground-truth image [see Fig. 7(c)] is manually sketched from the Landsat 8 image with reference to the Google Earth high-resolution images. The spectral signatures of the airport and roof are shown in Fig. 7(d)–(e). It should be noted that the targets are recognizable in the false-color image of 2014 [see Fig. 7(b)], but they did not exist in 2000 [see Fig. 7(a)], so the targets' spectra in 2000 actually reveal the features of bare land.

The detection results are shown in Fig. 8. Visually, the single-temporal algorithms cannot suppress the background information as well as our proposed algorithms, and it is difficult to recognize the targets in the results of single-temporal algorithms. In Fig. 8(a)–(c), CE, SWCEM, and E-CEM have a problem in distinguishing the road pixels from the target pixels. Meanwhile, in Fig. 8(d)–(g), MTCEM, RMTCEM, SCEM, and WTACEM extract the road pixels [the area in the green box in Fig. 8(d)] and some bare land with high reflectance [the area in the red box in Fig. 8(d)]. We plot the spectrum of a road pixel [background 1, marked by the blue arrow in Fig. 7(b)] and a high-reflectance bare land pixel [background 2, marked by the cyan arrow in Fig. 7(b)] in Fig. 7(d) and (e) and can see that the spectra of background pixels are analogous to the spectra of targets in both 2000 and 2014. However, the road and high-reflectance bare land pixels can be well suppressed by our FTA-based methods [see Fig. 8(h)–(k)], and the distinction between the targets and background pixels is more obvious in the results of our methods than in the results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

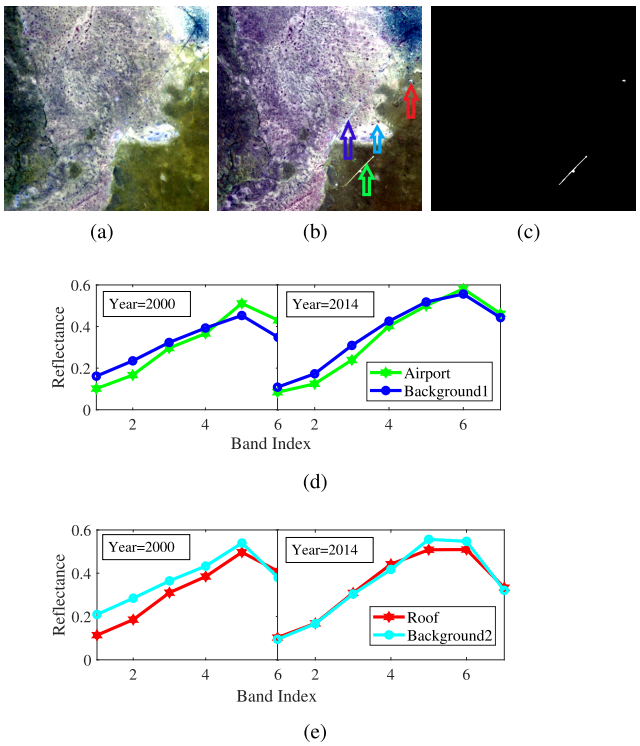


Fig. 7. Airport and roof data. (a) False-color image (R: band 4; G: band 3; and B: band 2) of 2000 Landsat 7 data. (b) False-color image (R: band 5; G: band 4; and B: band 3) of 2014 Landsat 8 data (the airport is marked by the green arrow; the roof is marked by the red arrow; the background marked by the blue arrow is called background 1; and the background marked by the cyan arrow is called background 2). (c) Ground-truth image. (d) Spectral signatures of the airport and background 1. (e) Spectral signatures of the roof and background 2.

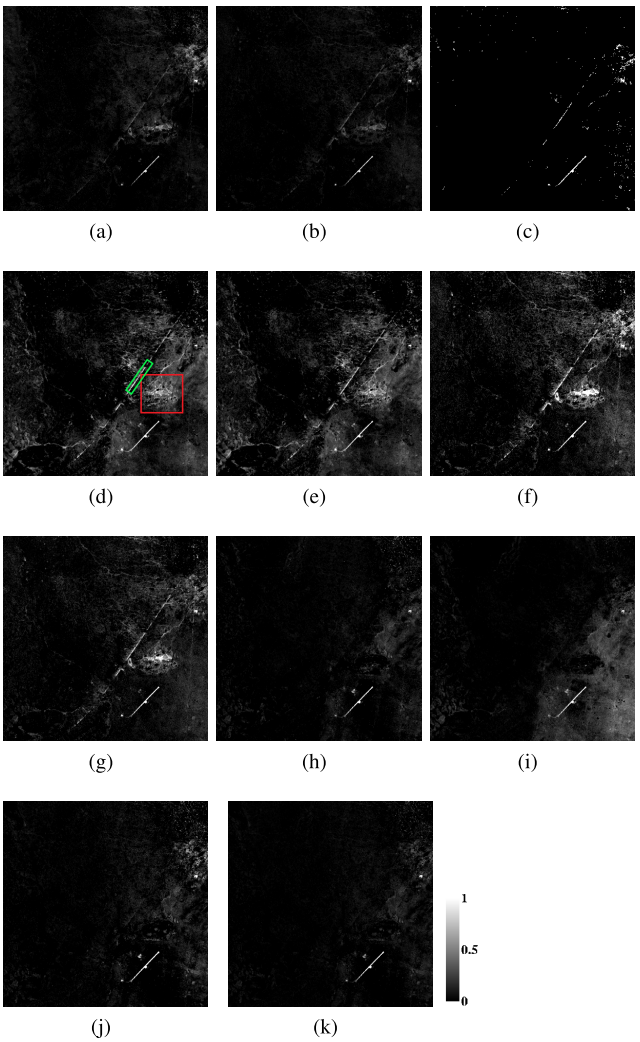


Fig. 8. Target detection results for the airport and roof data. (a) CE. (b) SWCEM. (c) E-CEM. (d) MTCEM (the pixels in the green box are the road pixels, and the pixels in the red box are the high-reflectance bare land pixels). (e) RMTCEM. (f) SCEM. (g) WTACEM. (h) MTFTA. (i) OWMTFTA. (j) SFTA. (k) WTAFTA.

of single-temporal methods. Among the proposed methods, WTAFTA can minimize the background interference visually.

The ROC curves of the detection results of different algorithms are plotted in Fig. 9, and the corresponding AUC metrics are tabulated in Table V. In Fig. 9, there exists an obvious gap between the ROC curves of our FTA-based (multitemporal) methods and of those single-temporal methods, indicating that our proposed methods work better in multitemporal multitarget detection tasks, so our methods reveal higher AUC values than the single-temporal methods. Furthermore, the accuracy metrics, OA, F-score, and kappa coefficient, are computed and tabulated in Table V. We can see that, although WTAFTA has the highest AUC value, SFTA gets the highest OA, F-score, and kappa coefficient. As discussed in the previous content, due to their different mechanisms, the two sets of accuracy metrics' evaluation of algorithms may be different. However, compared with those single-temporal methods, our multitemporal multitarget detection methods all have higher OA, F-score, and kappa coefficient.

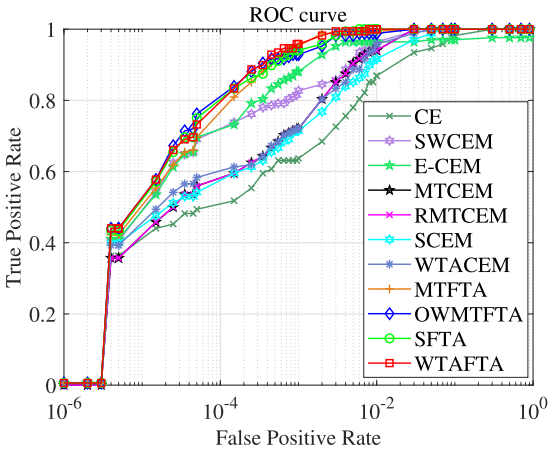


Fig. 9. ROC curves of the detection results of different algorithms for the airport and roof data. The x-axis is plotted in the logarithmic scale.

V. Discussion

In this section, we mainly discuss: 1) the algorithm complexity of our FTA-based algorithms and the CEM-based multitarget algorithms in Section II and 2) the performance of our FTA-based algorithms on the single-target detection problem in which the target spectrum has several variations.

A. Complexity Analysis

For a multitemporal data $\mathbf{X}^{(t)} = \{\mathbf{x}_1^{(t)}, \mathbf{x}_2^{(t)}, \ldots, \mathbf{x}_N^{(t)}\}$ (where $t = 1, 2, \ldots, M$) with q targets, the corresponding number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE V

AUC, OA, F-SCORE, AND KAPPA COEFFICIENT OF DIFFERENT ALGORITHMS FOR THE AIRPORT AND ROOF DATA. BOLD NUMBERS INDICATE THE HIGHEST VALUE IN EACH ROW

	CE	SWCEM	E-CEM	MTCEM	RMTCEM	SCEM	WTACEM	MTFTA	OWMTFTA	SFTA	WTAFTA
AUC	0.9908	0.9982	0.9748	0.9982	0.9982	0.9969	0.9978	0.9996	0.9996	0.9997	**0.9998**
OA	0.9560	0.9676	0.9836	0.9825	0.9825	0.9714	0.9773	0.9947	0.9909	**0.9971**	0.9950
F-score	0.9341	0.9516	0.9750	0.9740	0.9740	0.9579	0.9666	0.9921	0.9864	**0.9957**	0.9925
Kappa coefficient	0.9104	0.9512	0.9550	0.9643	0.9643	0.9165	0.9599	0.9866	0.9732	**0.9955**	0.9910

TABLE VI

COMPUTATIONAL COMPLEXITY OF MTCEM, RMTCEM, SCEM, WTACEM, MTFTA, OWMTFTA, SFTA, AND WTAFTA

Algorithm	Computational complexity
MTCEM	$O\left((L_1+L_2+\cdots+L_M)^3+(N+q)(L_1+L_2+\cdots+L_M)^2\right)$
RMTCEM	$O\left(2(L_1+L_2+\cdots+L_M)^3+[N+3q](L_1+L_2+\cdots+L_M)^2\right)$
SCEM/WTACEM	$O\left((L_1+L_2+\cdots+L_M)^3+(N+q)(L_1+L_2+\cdots+L_M)^2\right)$
MTFTA	$O\left((L_1\times L_2\times\cdots\times L_M)^3+(N+q)(L_1\times L_2\times\cdots\times L_M)^2\right)$
OWMTFTA	$O\left((L_1\times L_2\times\cdots\times L_M)^3+(2N+2q)(L_1\times L_2\times\cdots\times L_M)^2\right)$
SFTA/WTAFTA	$O\left((L_1\times L_2\times\cdots\times L_M)^3+(N+q)(L_1\times L_2\times\cdots\times L_M)^2\right)$

of bands for each time phase is $L_1, L_2, \ldots, L_M$, respectively. First, we analyze the computational complexity of MTCEM, RMTCEM, SCEM, WTACEM, MTFTA, SFTA, WTAFTA, and OWMTFTA, which are tabulated in Table VI.

Let $L = L_1 \times L_2 \times \cdots \times L_M$, and we use the derivation of the computational complexity of OWMTFTA (31) and (32) to illustrate how we get the expressions in Table VI. It mainly includes the following parts.

1) *Computation of* $\hat{\mathbf{r}}_i$*:* $O(NL)$.
2) *Computation of* $(\mathbf{I}-\mathbf{DD}^+)$*:* $O\left(2Lq^2+q^3+L^2q\right)$.
3) *Computation of* $\hat{\mathbf{R}}^*$*:* $O\left(N\left(2L^2+L\right)\right)$.
4) *computation of* $\hat{\mathbf{R}}^{*-1}$*:* $O\left(L^3\right)$.
5) *Computation of* $\hat{\mathbf{w}}_{\text{OWMTFTA}}$*:* $O\left(L^2q+Lq^2\right)$.

In practical applications, the number of bands is generally far greater than the number of targets, i.e., $L \gg q$. Then, the computational complexity of $(\mathbf{I}-\mathbf{DD}^+)$ can be simplified as $O\left(2Lq^2+L^2q\right)$.

Therefore, the total computational complexity of OWMTFTA is $O\left(L^3+(2N+2q)L^2+\left(2N+3q^2\right)L\right)$. Dropping the primary terms of L and substituting L for $L_1 \times L_2 \times \cdots \times L_M$, we get

$$O(_{\text{OWMTFTA}}) = O\left((L_1\times L_2\times\cdots\times L_M)^3 + (2N+2q)(L_1\times L_2\times\cdots\times L_M)^2\right). \quad (41)$$

From Table VI, we can see that the computational complexity of those methods is related to the following variables: the number of time phases (M), the number of bands in each time phase (L_i, $i = 1, \ldots, M$), the number of pixels (N), and the number of targets (q). In order to further explore the influence of those variables, the computational complexities of the FTA-based algorithms and CEM-based algorithms versus

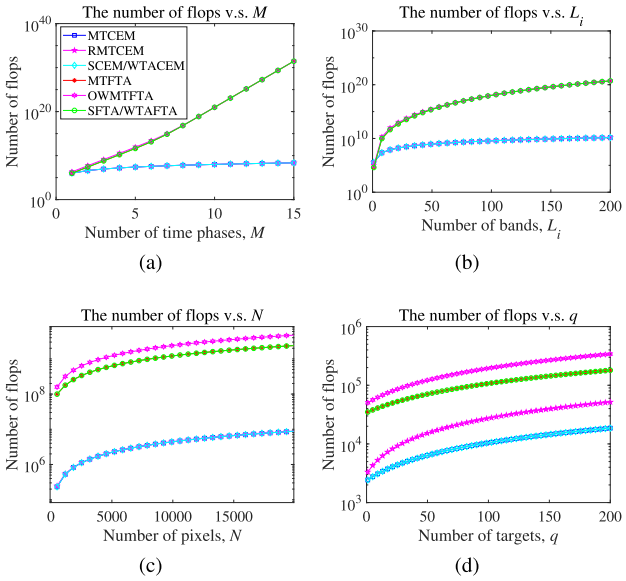


Fig. 10. Figures of the number of flops versus (a) number of time phases (M), (b) number of bands (L_i, suppose $L_i = L_1 = L_2 = \cdots = L_M$), (c) number of pixels (N), and (d) number of targets (q). The y-axis is plotted in the logarithmic scale.

M, L_i (suppose that $L_i = L_1 = L_2 = \cdots = L_M$), N, and q are plotted in Fig. 10. It can be seen that the FTA-based methods have higher computational complexity than the CEM-based methods. Moreover, the computational complexity of both FTA- and CEM-based methods increases with M, L_i, N, and q. Especially, the complexity of FTA-based methods grows much more rapidly than that of CEM-based methods as M increases.

Next, we discuss the spatial complexity of FTA-based methods. From (28) and (32)–(34), we see the spatial complexity of MTFTA, OWMTFTA, SFTA, and WTAFTA mainly depends on the dimension of the autocorrelation matrix $\hat{\mathbf{R}}$, whose size is $(L_1 \times L_2 \times \cdots \times L_M) \times (L_1 \times L_2 \times \cdots \times L_M)$. Thus, FTA-based algorithms require a very large computational memory for multitemporal data with a large number of spectral bands. Our future work will focus on how to reduce the memory requirement of FTA-based methods.

B. Single-Target Detection Using the Proposed Methods

We discuss the effectiveness of the multitarget detection methods (i.e., FTA-based methods) and of the single-target detection method (i.e., FTA) on the single-target detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

ACCURACY METRICS, NAMELY, AUC, OA, F-SCORE, AND KAPPA COEFFICIENT OF FTAS, MTFTA, OWMTFTA, SFTA, AND WTAFTA FOR THE SINGLE-TARGET EXPERIMENT. BOLD NUMBERS INDICATE THE HIGHEST VALUE IN EACH ROW

	FTA 1	FTA 2	FTA 3	FTA 4	MTFTA	OWMTFTA	SFTA	WTAFTA
AUC	0.5980	0.5648	0.5722	0.6440	0.7372	**0.8790**	0.7281	0.8216
OA	0.6151	0.6134	0.5931	0.6681	0.7203	**0.8476**	0.7251	0.7706
F-score	0.5267	0.5060	0.5035	0.5869	0.6673	**0.8198**	0.6660	0.7373
kappa coefficient	0.2043	0.1935	0.1603	0.3118	0.4263	**0.6879**	0.4332	0.5339

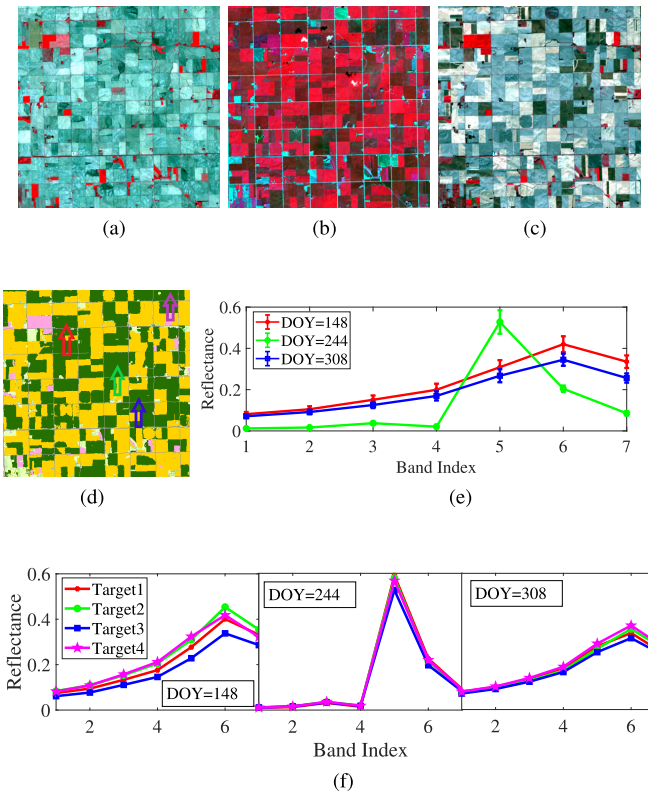


Fig. 11. Farmland data (Landsat 8, path = 029, and row = 031) in year 2014. (a) False-color image (R: band 5; G: band 4; and B: band 3) with DOY = 148. (b) False-color image with DOY = 244. (c) False-color image with DOY = 308. (d) Ground-truth image derived from 2014 CDL data (dark green: soybeans (target 1 is marked by the red arrow; target 2 is marked by the green arrow; target 3 is marked by the blue arrow; and target 4 is marked by the purple arrow), yellow: corn, lemon chiffon: grassland, light green: peas, and magenta: alfalfa). (e) Target signatures (mean ± standard deviation) of soybeans. (f) Spectra of targets from four locations.

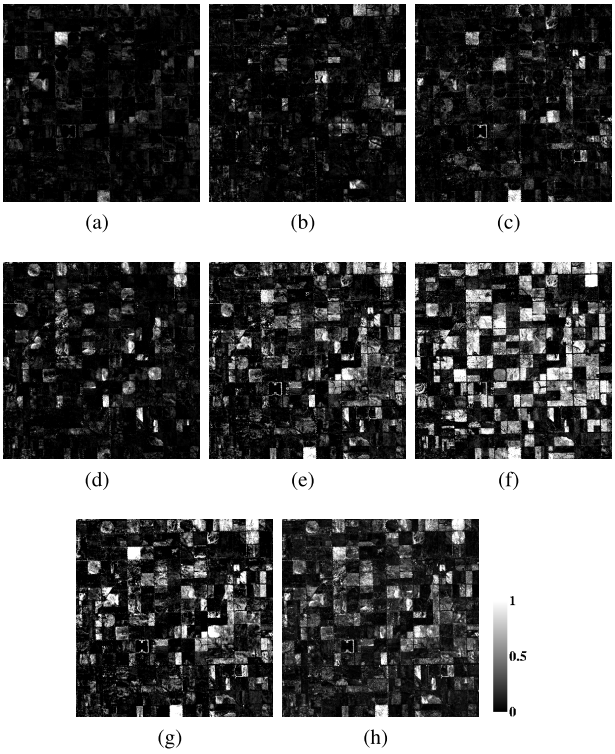


Fig. 12. Target detection results for the farmland data. (a) FTA 1. (b) FTA 2. (c) FTA 3. (d) FTA 4. (e) MTFTA. (f) OWMTFTA. (g) SFTA. (h) WTAFTA.

problem. Even for a field covered by the same ground object, the spectra from different locations may hold some differences from one another. In order to illustrate this situation, we still use the farmland data in Section IV-B. Yet, the data used here are of a dimension of 400 × 400 × 7 and contain three time phases (DOY = 148, 244, and 308, respectively). The false-color images are displayed in Fig. 11(a)–(c). The ground-truth image [see Fig. 11(d)] is also obtained from the 2014 CDL data with reference to Landsat 8 multispectral images. The soybeans [the areas in green in Fig. 11(d)] are selected as the targets of interest, which occupies a large area in the image. The spectral signatures (mean ± standard deviation) of the soybeans in three time phases are plotted in Fig. 11(e). We can observe that the spectra of soybeans in DOY = 148 and DOY = 308 have very large deviations. Thus, one single spectrum cannot fully represent the spectra of all soybeans, which is the reason why we select four spectra from different parts of the image as the target spectra. The targets are marked with arrows in different colors in the ground-truth image [see Fig. 11(d)] (target 1 is marked by the red arrow; target 2 is marked by the green arrow; target 3 is marked by the blue arrow; and target 4 is marked by the purple arrow). The spectra of the targets are plotted in Fig. 11(f).

First, we detect each of the target signatures through FTA separately, and the detection results are called FTA 1–FTA 4, respectively, as shown in Fig. 12(a)–(d). Then, we put together the four target signatures and apply the multitemporal multitarget algorithms, whose detection results are displayed in Fig. 12(e)–(h). Visually, the multitarget detection algorithms for multitemporal data have better detection results than the single-target detection algorithm for multitemporal data. We can see that FTAs for a single target can only detect a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

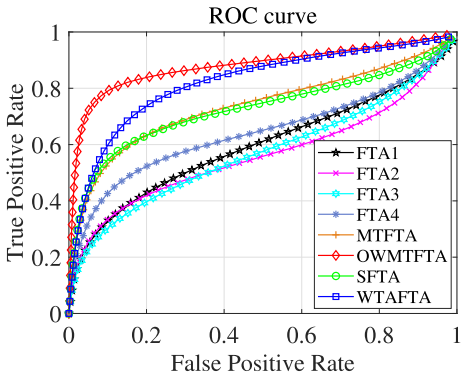


Fig. 13. ROC curves of the detection results of FTAs, MTFTA, OWMTFTA, SFTA, and WTAFTA for the single-target experiment.

part of the target area [see Fig. 12(a)–(d)], which is because the spectra of soybeans vary from location to location and FTA is sensitive to the change of target signature. In contrast, FTA-based algorithms, MTFTA, OWMTFTA, SFTA, and WTAFTA [see Fig. 12(e)–(h)], have displayed better detection results than the FTAs, especially OWMTFTA has the best performance.

In order to compare the accuracy of those detection algorithms quantitatively, their ROC curves are plotted in Fig. 13, and the AUC metrics are tabulated in Table VII. It is seen that the multitarget algorithms for multitemporal data, MTFTA, OWMTFTA, SFTA, and WTAFTA, outperform the single-target detection algorithm (i.e., FTA). In particular, the detection result of OWMTFTA has the best ROC curve. It is because the soybean occupies a large part of the image, i.e., the soybean is a "big" target in this image. Thereby, applying the weighted autocorrelation matrix rather than the original one can better suppress the background. Moreover, OA, F-score, and kappa coefficient of the detection results (see Table VII) show that the multitarget algorithms for multitemporal data can achieve better binary classification than the single-target detection algorithm for multitemporal data in the single-target experiment. Among those methods, OWMTFTA has the best performance, which coincides with our analysis mentioned above.

VI. Conclusion

In the field of hyper/multispectral remote sensing image processing, the algorithms for multitarget detection in multitemporal data are needed intensely. In this article, we present several multitarget detection algorithms for multitemporal remote sensing data, called MTFTA, OWMTFTA, SFTA, and WTAFTA. They are derived from FTA with the same patterns that the multitarget algorithms for single-temporal data were derived from CEM. A series of experiments are conducted on different data sets with our proposed algorithms and some single-temporal algorithms, and the experiment results illustrate that our FTA-based methods show more competitive performance compared with other target detection algorithms for single-temporal data. Future works include MTFTA, OWMTFTA, SFTA, and WTAFTA's applications on hyperspectral images by reducing the computational complexity of the algorithms and also include finding the best application situations for each of the four algorithms.

References

[1] F. Liu, L. Jiao, X. Tang, S. Yang, W. Ma, and B. Hou, "Local restricted convolutional neural network for change detection in polarimetric SAR images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 3, pp. 818–833, Mar. 2019.

[2] X. Tang, C. Liu, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Large-scale remote sensing image retrieval based on semi-supervised adversarial hashing," *Remote Sens.*, vol. 11, no. 17, p. 2055, Sep. 2019.

[3] X. Zhang, X. Wang, X. Tang, H. Zhou, and C. Li, "Description generation for remote sensing images using attribute attention mechanism," *Remote Sens.*, vol. 11, no. 6, p. 612, Mar. 2019.

[4] D. Manolakis *et al.*, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[5] W. Liu, F. Yamazaki, and T. T. Vu, "Automated vehicle extraction and speed determination from QuickBird satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 1, pp. 75–82, Mar. 2011.

[6] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[7] D. Manolakis, C. Siracusa, and G. Shaw, "Hyperspectral subpixel target detection using the linear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1392–1409, Jul. 2001.

[8] J. P. Kerekes and J. E. Baum, "Spectral imaging system analytical model for subpixel object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 5, pp. 1088–1101, May 2002.

[9] M. S. Stefanou and J. P. Kerekes, "Image-derived prediction of spectral image utility for target detection applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 4, pp. 1827–1833, Apr. 2010.

[10] X. Li, S. Zhang, X. Pan, P. Dale, and R. Cropp, "Straight road edge detection from high-resolution remote sensing images based on the ridgelet transform with the revised parallel-beam radon transform," *Int. J. Remote Sens.*, vol. 31, no. 19, pp. 5041–5059, Oct. 2010.

[11] J. Han *et al.*, "Efficient, simultaneous detection of multi-class geospatial targets based on visual saliency modeling and discriminative learning of sparse coding," *ISPRS J. Photogramm. Remote Sens.*, vol. 89, pp. 37–48, Mar. 2014.

[12] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," *Acta Ecologica Sinica*, vol. 28, no. 2, pp. 627–635, 2015.

[13] Y. Chen, X. Chen, J. Zhou, Y. Ji, and W. Shen, "Camouflage target detection via hyperspectral imaging plus information divergence measurement," in *Proc. Int. Conf. Optoelectronics Microelectron. Technol. Appl.*, Jan. 2017, Art. no. 102440F.

[14] E. K. Douaoui, H. Nicolas, and C. Walter, "Detecting salinity hazards within a semiarid context by means of combining soil and remote-sensing data," *Geoderma*, vol. 134, no. 1, pp. 217–230, 2017.

[15] C.-I. Chang and D. C. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.

[16] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[17] D. Chaudhuri, N. K. Kushwaha, and A. Samal, "Semi-automated road detection from high resolution satellite images by directional morphological enhancement and segmentation techniques," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 5, pp. 1538–1544, Oct. 2012.

[18] J. Zhang, X. Lin, Z. Liu, and J. Shen, "Semi-automatic road tracking by template matching and distance transformation in urban areas," *Int. J. Remote Sens.*, vol. 32, no. 23, pp. 8331–8347, Dec. 2011.

[19] G. Liu, X. Sun, K. Fu, and H. Wang, "Interactive geospatial object extraction in high resolution remote sensing images using shape-based global minimization active contour model," *Pattern Recognit. Lett.*, vol. 34, no. 10, pp. 1186–1195, Jul. 2013.

[20] C. Hung, M. Bryson, and S. Sukkarieh, "Multi-class predictive template for tree crown detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 68, pp. 170–183, Mar. 2012.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.

[21] S. Leninisha and K. Vani, "Water flow based geometric active deformable model for road network," *ISPRS J. Photogramm. Remote Sens.*, vol. 102, pp. 140–147, Apr. 2015.

[22] A. O. Ok, C. Senaras, and B. Yuksel, "Automated detection of arbitrarily shaped buildings in complex environments from monocular VHR optical satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1701–1717, Mar. 2013.

[23] X. Li and G. Shao, "Object-based urban vegetation mapping with high-resolution aerial photography as a single data source," *Int. J. Remote Sens.*, vol. 34, no. 3, pp. 771–789, Feb. 2013.

[24] J. Im, L. J. Quackenbush, M. Li, and F. Fang, "Optimum scale in object-based image analysis," in *Proc. Scale Issues Remote Sens.*, 2014, pp. 197–214.

[25] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, vol. 1, no. 1, pp. 886–893.

[26] F.-F. Li and P. Perona, "A Bayesian hierarchical model for learning natural scene categories," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2005, pp. 524–531.

[27] F.-F. Li and P. Perona, "Statistical learning theory," in *Learning From Data*. Hoboken, NJ, USA: Wiley, 2011, pp. 99–150.

[28] G. Hu, Z. Yang, J. Han, L. Huang, J. Gong, and N. Xiong, "Aircraft detection in remote sensing images based on saliency and convolution neural network," *EURASIP J. Wireless Commun. Netw.*, vol. 2018, no. 1, pp. 1–6, Feb. 2018.

[29] Y. Feng, L. Wang, and M. Zhang, "A multi-scale target detection method for optical remote sensing images," *Multimedia Tools Appl.*, vol. 78, no. 7, pp. 8751–8766, Jul. 2018.

[30] Q. Guo, S. Li, Q. Wang, Z. Hu, and Q. Jing, "Research of target detection algorithm based on the remote sensing image," in *Proc. Signal Inf. Process., Netw. Comput.*, Jan. 2020, pp. 369–375.

[31] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, early access, May 18, 2020, doi: 10.1109/TGRS.2020.2991407.

[32] G. Cheng, J. Yang, D. Gao, L. Guo, and J. Han, "High-quality proposals for weakly supervised object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 5794–5804, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9069411, doi: 10.1109/TIP.2020.2987161.

[33] X. Feng, J. Han, X. Yao, and G. Cheng, "Progressive contextual instance refinement for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8002–8012, Nov. 2020.

[34] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Dec. 2002.

[35] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 427–438, Feb. 2005.

[36] D. Manolakis, "Is there a best hyperspectral detection algorithm?" *Proc. SPIE Newsroom*, vol. 7334, Apr. 2009, Art. no. 733402.

[37] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Comput. Sci. Elect. Eng., Univ. Maryland Baltimore County, Baltimore, MD, USA, Jan. 1993.

[38] X. Geng, L. Ji, and K. Sun, "Clever eye algorithm for target detection of remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 32–39, Apr. 2016.

[39] T. Wang, B. Du, and L. Zhang, "A kernel-based target-constrained interference-minimized filter for hyperspectral sub-pixel target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 626–637, Apr. 2013.

[40] H. Kwon and N. M. Nasrabadi, "Kernel spectral matched filter for hyperspectral imagery," *Int. J. Comput. Vis.*, vol. 71, no. 2, pp. 127–141, Feb. 2007.

[41] X. Geng, W. Yang, L. Ji, C. Ling, and S. Yang, "A piecewise linear strategy of target detection for multispectral/hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 951–961, Mar. 2018.

[42] S. Yang and Z. Shi, "SparseCEM and SparseACE for hyperspectral image target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 12, pp. 2135–2139, Dec. 2014.

[43] X. Geng, W. Yang, L. Ji, F. Wang, and Y. Zhao, "The match filter (MF) is always superior to constrained energy minimization (CEM)," *Remote Sens. Lett.*, vol. 8, no. 7, pp. 696–702, Jul. 2017.

[44] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.

[45] S. Yang, Z. Shi, and W. Tang, "Robust hyperspectral image target detection using an inequality constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3389–3404, Jun. 2015.

[46] Y. Wang, S. Huang, D. Liu, and H. Wang, "A target detection method for hyperspectral imagery based on two-time detection," *J. Indian Soc. Remote Sens.*, vol. 45, no. 2, pp. 239–246, Apr. 2017.

[47] Y. Wang, M. Fan, J. Li, and Z. Cui, "Sparse weighted constrained energy minimization for accurate remote sensing image target detection," *Remote Sens.*, vol. 9, no. 11, p. 1190, Nov. 2017.

[48] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, 2019.

[49] J. Du and Z. Li, "A hyperspectral target detection framework with subtraction pixel pair features," *IEEE Access*, vol. 6, pp. 45562–45577, 2018.

[50] W. Xie, J. Yang, J. Lei, Y. Li, Q. Du, and G. He, "SRUN: Spectral regularized unsupervised networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1463–1474, Feb. 2020.

[51] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 29, 2020, doi: 10.1109/TGRS.2020.3001948.

[52] Y. Shi, J. Li, Y. Yin, B. Xi, and Y. Li, "Hyperspectral target detection with macro-micro feature extracted by 3-D residual autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4907–4919, Dec. 2019.

[53] Y. Shi, J. Lei, Y. Yin, K. Cao, Y. Li, and C.-I. Chang, "Discriminative feature learning with distance constrained stacked sparse autoencoder for hyperspectral target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1462–1466, Sep. 2019.

[54] H. Ren, Q. Du, C.-I. Chang, and J. O. Jensen, "Comparison between constrained energy minimization based approaches for hyperspectral imagery," in *Proc. IEEE Workshop Adv. Techn. Anal. Remotely Sensed Data*, Oct. 2003, pp. 244–248.

[55] X. Xiao *et al.*, "Mapping paddy rice agriculture in southern China using multi-temporal MODIS images," *Remote Sens. Environ.*, vol. 95, no. 4, pp. 480–492, Apr. 2005.

[56] S.-K. Ning, N.-B. Chang, K.-Y. Jeng, and Y.-H. Tseng, "Soil erosion and non-point source pollution impacts assessment with the aid of multi-temporal remote sensing images," *J. Environ. Manage.*, vol. 79, no. 1, pp. 88–101, Apr. 2006.

[57] S.-B. Gao, Y.-M. Cheng, Y.-Q. Zhao, K. Wei, and Q. Pan, "Detection of buried targets based on multitemporal infrared image," *J. Infr. Millim.*, vol. 28, no. 1, pp. 25–30, Mar. 2009.

[58] C. R. Fichera, G. Modica, and M. Pollino, "Land cover classification and change-detection analysis using multi-temporal remote sensed imagery and landscape metrics," *Eur. J. Remote Sens.*, vol. 45, no. 1, pp. 1–18, Jan. 2012.

[59] X. Geng, L. Ji, and Y. Zhao, "Filter tensor analysis: A tool for multi-temporal remote sensing target detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 290–301, May 2019.

[60] X. Geng and Y. Zhao, "Principle of small target detection for hyperspectral imagery," *Sci. China Ser. D, Earth Sci.*, vol. 50, no. 8, pp. 1225–1231, Aug. 2007.

[61] X. Zhang, *Matrix Analysis and Applications*. Cambridge, U.K.: Cambridge Univ. Press, 2017.

[62] L. Ji, X. Geng, K. Sun, Y. Zhao, and P. Gong, "Target detection method for water mapping using Landsat 8 OLI/TIRS imagery," *Water*, vol. 7, no. 12, pp. 794–817, Feb. 2015.

[63] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.

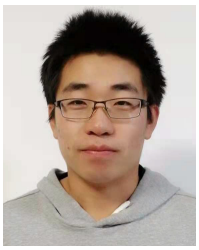
[64] L. Ji, P. Gong, J. Wang, J. Shi, and Z. Zhu, "Construction of the 500m resolution daily global surface water change database (2001–2016)," *Water Resour. Res.*, vol. 54, no. 12, p. 270, Dec. 2018.

[65] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950.

[66] W. J. Youden, "The statistical evaluation of medical tests for classification and prediction," *Technometrics*, vol. 47, no. 2, p. 245, May 2005.

[67] N. J. Perkins and E. F. Schisterman, "The inconsistency of 'optimal' cutpoints obtained using two criteria based on the receiver operating characteristic curve," *Amer. J. Epidemiol.*, vol. 163, no. 7, pp. 670–675, Jan. 2006.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.



Yanxin Xi received the B.S. degree in automation from the University of Science and Technology of China, Hefei, China, in 2017, and the M.S. degree in signal processing from the Key Laboratory of Technology in Geo-spatial Information processing and Application Systems, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2020.

His research interests include hyperspectral target detection and machine learning.



Luyan Ji received the B.S. degree in electrical engineering from the University of Science and Technology of China, Hefei, China, in 2008, and the Ph.D. degree in signal processing from the Institute of Electronics, Chinese Academic of Sciences, Beijing, China, in 2013.

She is an Assistant Professor with the Key Laboratory of Technology in the Geo-spatial information processing and application systems, Institute of Electronics, Chinese Academy of Sciences. Her research interests include issues related to the spectral–temporal characteristics of soil water-holding capacity, the water mapping, spectral mixing analysis, target detection, and feature extraction in hyperspectral remote sensing application.



Weitun Yang received the B.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2014, and the M.S. degree in signal processing from the Institute of Electronics, Chinese Academic of Sciences, Beijing, in 2017. He received his Ph.D. degree in signal and information processing from the Key Laboratory of Technology in Geo-spatial Information Processing and Application Systems, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, in 2020.

His research interests include hyperspectral target detection, image registration, and machine learning.



Xiurui Geng received the bachelor's and master's degrees in applied mathematics from Beihang University, Beijing, China, in 1999 and 2002, respectively, and the Ph.D. degree in hyperspectral remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, in 2005.

He is a Professor with the Key Laboratory of Technology in Geo-spatial Information processing and Application Systems, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests are hyperspectral data understanding and algorithm development in feature extraction, mixed pixel analysis, image matching, target detection, and classification by introducing new algebra, geometry, analysis, and statistics viewpoints.



Yongchao Zhao received the Ph.D. degree in geology from Peking University, Beijing, China, in 1999.

He is a Professor with the Key Laboratory of Technology in Geo-spatial Information processing and Application Systems, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include hyperspectral image precise processing, including calibration, preprocessing, and optimization; spectral characteristics of typical terrestrial objects; integrative apparent spectral models of these typical objects; and information extraction from hyperspectral images' data based on the precise understanding of spectral characteristics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 06:51:00 UTC from IEEE Xplore. Restrictions apply.